

- 15** Influenza virus has an enzyme called neuraminidase which breaks down glycoproteins in the surface membrane of the cell that the virus will infect. The glycoprotein binds to the active site of neuraminidase by induced fit.

Which statements about the induced fit hypothesis of enzyme action are correct?

- 1 The active site must have the same shape as the substrate for them to bind together.
- 2 This enzyme is less likely to be affected by non-competitive inhibitors than an enzyme working by the lock-and-key mechanism.
- 3 The substrate is converted to product by specific R-groups in the active site just like the lock-and-key mechanism.

**A** 1 and 2

**B** 2 and 3

**C** 2 only

**D** 3 only